

TABLE 3-4A
LVSHIM SPECIFICATIONS – BRM AND TRM(WB)

LVSHIM TYPE	OVERALL HOMOGENEITY		HARMONIC COEFFICIENTS (ZERO TO PEAK)			
	"SHIM TO" GOAL*	CUSTOMER ACCEPTANCE	MAGNET TYPE	HARMONIC NAME	"SHIM TO" GOAL*	CUSTOMER ACCEPTANCE
Main (45-cm DSV 500Hz BW)	<45 Hz	< 80 Hz after Gradshim	Oxford (12 coils)	Z5, Z6, Z(X2-Y2), ZXY, X3, Y3 Z3, Z4, Z2X, Z2Y Z2, ZX, ZY, X2-Y2, XY	NA** < 100 mA† delta current < ±25 Hz	NA NA < ±50 Hz after Gradshim
Main (45-cm DSV 500Hz BW)	<45 Hz	< 80 Hz after Gradshim	S-I (14 coils)	Z5, Z6, X3, Y3 Z3, Z4, Z2X, Z2Y Z2, ZX, ZY, X2-Y2, XY X, Y, Z1	NA** < 100 mA† delta current < ±25 Hz < ±10 Hz	NA NA < ±50 Hz after Gradshim NA
Main (45-cm DSV 500Hz BW)	<45 Hz	< 80 Hz after Gradshim	S-II, S-III, S-IV, S-X, S-XC_1 (18 coils)	Z6 Z3, Z4, Z5, Z2X, Z2Y, Z(X2-Y2), ZXY, X3, Y3 Z2, ZX, ZY, X2-Y2, XY X, Y, Z1	< ±200 Hz‡ < 100 mA† delta current < ±25 Hz < ±10 Hz	NA NA < ±50 Hz after Gradshim NA
Main (45-cm DSV 500Hz BW)	<45 Hz	< 80 Hz after Gradshim	S-V, S-XC_2 (12 coils)	X2-Y2, XY, Z(X2-Y2), ZXY, X3, Y3 Z6 Z3, Z4, Z5, Z2X, Z2Y Z2, ZX, ZY X, Y, Z1	NA** < ±200 Hz‡ < 100 mA† delta current < ±25 Hz < ±10 Hz	NA NA NA < ±50 Hz after Gradshim NA
Gradient (22-cm DSV)	< 6 Hz	< 12 Hz after Gradshim	All , except CX & LCC magnets (3 coils)	X, Y Z1	≥ ±1 delta gradient unit	< 200 total gradient units

* Most magnets should be able to achieve the "Shim To" values shown. All magnets must meet the "Customer Acceptance" values. As long as the magnet meets the Customer Acceptance values, Gradient LVshim should be sufficient to maintain an acceptable shim quality. When the Customer Acceptance values are exceeded, Main LVshim must be performed to re-establish an acceptable shim quality.

** Oxford, S-I, S-V, and S-XC_2 magnets do not have a full set of 18 shim coils; therefore, the harmonics associated with the "missing" coils do not have LVshim specifications. Passive shimming is required to reduce these harmonics. For S-V and S-XC_2 magnets, if X2-Y2, or XY, are > ±25 Hz, you must perform passive to reduce these to < ±25 Hz.

† Note that this spec is for < 100 mA delta current, **NOT** < ±100 Hz. As a rule of thumb, all the power supply delta currents should be below 100 mA.

‡ The specification for the Z6 harmonic can be ignored if it forces the center frequency to move out of specification (i.e., forcing the magnet to be re-parked) or if it forces the current in the Z4 or Z6 power supply to exceed 20 amps.

TABLE 3-4A (Continued)
LVSHIM SPECIFICATIONS – BRM AND TRM(WB)

LVSHIM TYPE	OVERALL HOMOGENEITY		HARMONIC COEFFICIENTS (ZERO TO PEAK)			
	"SHIM TO" GOAL*	CUSTOMER ACCEPTANCE	MAGNET TYPE	HARMONIC NAME	"SHIM TO" GOAL*	CUSTOMER ACCEPTANCE
Main LVshim 45 cm dsv 6 scan Planes 500 Hz BW LVshim 50 cm Diam Phantom	< 33 Hz < 20 units of Gradshim	< 54 Hz after Gradshim	Cx 1.0T & LCC 1.0T (18 coils)	1, 0 (Z1)	< 15 Hz	see Gradshim
				2, 0 (Z2)	< 25 Hz	< 50 Hz
				3, 0 (Z3)	< 25 Hz	< 50 Hz
				4, 0 (Z4)	< 50 Hz	< 100 Hz
				5, 0 (Z5)	< 50 Hz	< 100 Hz
				6, 0 (Z6)	< 50 Hz	< 100 Hz
				8, 0 (Z8)	No spec	No spec
				10, 0 (Z10)	No spec	No spec
				12, 0 (Z12)	No spec	No spec
				1, 1 (Y)	< 15 Hz	see Gradshim
				1, -1 (X)	< 15 Hz	see Gradshim
				2, 1 (ZY)	< 25 Hz	< 50 Hz
				2, -1 (ZX)	< 25 Hz	< 50 Hz
				2, 2 (X2-Y2)	< 25 Hz	< 50 Hz
				3, 1 (Z2Y)	< 25 Hz	< 50 Hz
				3, -1 (Z2X)	< 25 Hz	< 50 Hz
				3, 2 (ZX2-ZY2)	< 25 Hz	< 50 Hz
				3, -2 (ZXY)	< 25 Hz	< 50 Hz
				3, 3 (Y3)	< 25 Hz	< 50 Hz
				3, -3 (Y3)	< 25 Hz	< 50 Hz
Gradshim 22cm DSV	< 4 Hz	< 6 Hz after Gradshim		XYZ	≤ 1unit	< 200 units
6 scan planes, 200 Hz BW, TLT or LVshim Phantom						
Main LVshim 45 cm dsv 6 scan Planes 500 Hz BW LVshim 50 cm Diam Phantom	< 50 Hz < 20 units of Gradshim	< 80 Hz after Gradshim	Cx 1.5T & LCC 1.5T (18 coils)	1, 0 (Z1)	< 10 Hz	see Gradshim
				2, 0 (Z2)	< 25 Hz	< 50 Hz
				3, 0 (Z3)	< 25 Hz	< 50 Hz
				4, 0 (Z4)	< 50 Hz	< 100 Hz
				5, 0 (Z5)	< 50 Hz	< 100 Hz
				6, 0 (Z6)	< 50 Hz	< 100 Hz
				8, 0 (Z8)	No spec	No spec
				10, 0 (Z10)	No spec	No spec
				12, 0 (Z12)	No spec	No spec
				1, 1 (Y)	< 10 Hz	see Gradshim
				1, -1 (X)	< 10 Hz	see Gradshim
				2, 1 (ZY)	< 25 Hz	< 50 Hz
				2, -1 (ZX)	< 25 Hz	< 50 Hz
				2, 2 (X2-Y2)	< 25 Hz	< 50 Hz
				3, 1 (Z2Y)	< 25 Hz	< 50 Hz
				3, -1 (Z2X)	< 25 Hz	< 50 Hz
				3, 2 (ZX2-ZY2)	< 25 Hz	< 50 Hz
				3, -2 (ZXY)	< 25 Hz	< 50 Hz
				3, 3 (Y3)	< 25 Hz	< 50 Hz
				3, -3 (Y3)	< 25 Hz	< 50 Hz
Gradshim 22cm DSV	< 4 Hz	< 6 Hz after Gradshim	All CX & LCC	XYZ	≤ 1unit	< 200 units
6 scan planes, 200 Hz BW, TLT or LVshim Phantom						